

2D Plotting Using the *matplotlib* Library

Publication quality 2D plots can be produced by *matplotlib*, which is an open source object-oriented Python library. With this article, we begin a series that will take the reader through the nuances of 2D plotting with *matplotlib*.

matplotlib is a Python library for creating 2D plots. It was originally created by John D. Hunter and is now maintained by a large team of developers. It integrates well with *IPython* – the component in the standard scientific Python toolset. *matplotlib* uses NumPy – a fundamental package for scientific computing with Python. The *matplotlib* code is conceptually divided into three parts -- the *pylab* interface, the front-end and the back-end. The *pylab* interface is a set of functions which allow the user to create plots. The front-end (*matplotlib* API) is a set of classes for creating and managing figures, text, lines, plots, etc. Basically, this is an abstract interface. The back-end is a drawing device (renderer) that transforms the front-end representation to a hard copy or a display device.

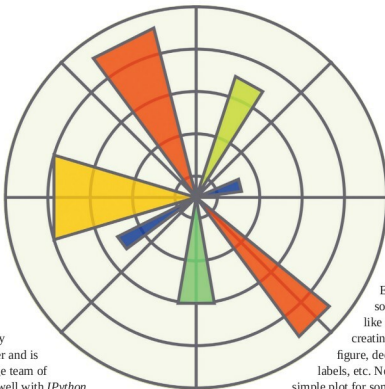
Installation

Use the following commands to install *matplotlib*:

```
python -m pip install -U pip setuptools
python -m pip install matplotlib
```

Pyplot

matplotlib.pyplot is a collection of command style functions.



below:

```
import matplotlib.pyplot as plt
plt.plot([1,2,3,4])
plt.ylabel('Y values')
plt.show()
```

If you provide a single list array to the *plot()* command, *matplotlib* assumes it is a sequence of Y values and internally generates the X value for you. The output is shown in Figure 1.

We do not need to worry about window creation and other basic event handling. *matplotlib* provides some basic operations of the plot. The basic operations are: *Reset original view*, *Back to previous view*, *Forward to next view*, *Pan axes with left mouse*, *Zoom with right*, *Zoom to rectangle*, *Configure subplots* and *Save the figure*. If you click on the 'Configure subplots' button, another window with the following parameters will open: *left*, *bottom*, *right*, *top*, *wspace*, and *hspace* to adjust the plot (see Figure 2).